Predicting Recession Risk Using Macroeconomic Indicators

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Background: Macroeconomics and Recession

- Economic Recession: A "significant decline in economic activity that is spread across the economy and that lasts more than a few months"
- Different from the ebb and flow of the economy at large
- These periods standout and are marked by diminishing GDP and several other econometrics
- Heavily researched topic in macroeconomics

Research Questions

- What are the drivers and reasons that a recession occurs?
  - What econometrics make the largest impact
- Recessions are often drastic and almost unique from the ordinary economic cycle, can we have our model capture their magnitude? (Student t-model)
- Probability of a recession for a quarter given the previous quarter's data (Frequentist and Bayesian Linear Regression)
- Influence of predictors on a quarter's GDP growth over time (Rolling Window Coefficient Estimation)

Data

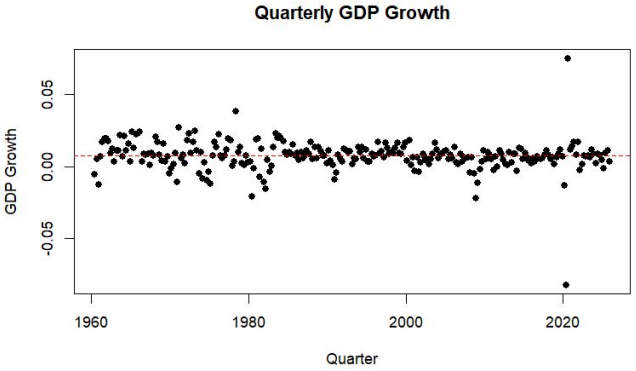
- Quarterly macroeconomic dataset (FRED-QD) from the Federal Reserve Bank of St. Louis
- The dataset contains over six decades worth of observed data on the quarterly level, and over 200 time-series variables
- Predictors: GDP growth, unemployment rate, inflation, industrial production growth, and treasury yield spread
- Response: GDP growth
- In our model we took the log difference of these predictors in order to get lagged predictors
- Our final dataset had 263 observations (Quarters) and 7 Variables (including date and response)

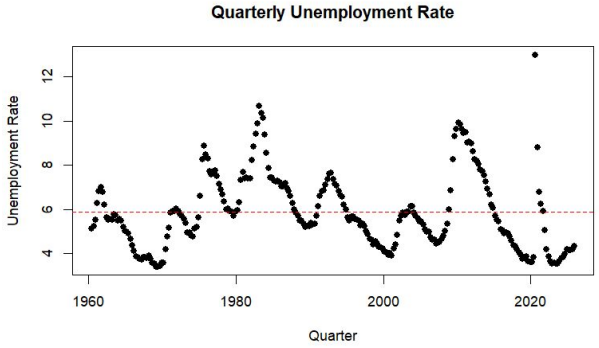
Data contd.

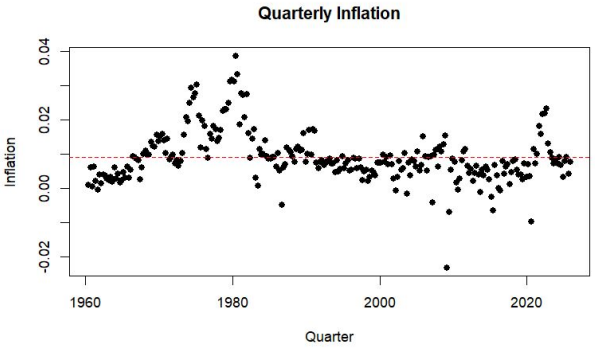
date	gdp_growth	lag_gdp_growth	lag_unrate
Min. :1960-06-01	Min. :-0.082043	Min. :-0.082043	Min. : 3.400
1st Qu.:1976-10-16	1st Qu.: 0.003493	1st Qu.: 0.003493	1st Qu.: 4.633
Median :1993-03-01	Median : 0.007602	Median : 0.007681	Median : 5.567
Mean :1993-03-01	Mean : 0.007320	Mean : 0.007391	Mean : 5.869
3rd Qu.:2009-07-17	3rd Qu.: 0.011239	3rd Qu.: 0.011331	3rd Qu.: 6.917
Max. :2025-12-01	Max. : 0.074758	Max. : 0.074758	Max. :13.000

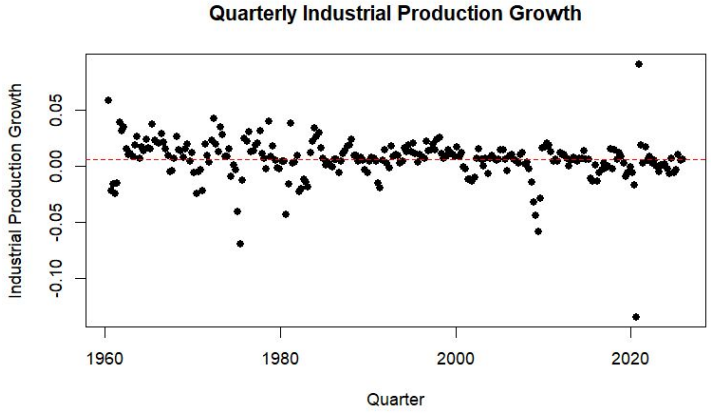
lag_inflation	lag_ip_growth	lag_spread
Min. :-0.023168	Min. :-0.1343803	Min. :-1.480
1st Qu.: 0.005033	1st Qu.:-0.0003963	1st Qu.: 0.435
Median : 0.007831	Median : 0.0067439	Median : 1.473
Mean : 0.009119	Mean : 0.0057158	Mean : 1.399
3rd Qu.: 0.011336	3rd Qu.: 0.0142108	3rd Qu.: 2.502
Max. : 0.038698	Max. : 0.0905126	Max. : 3.800

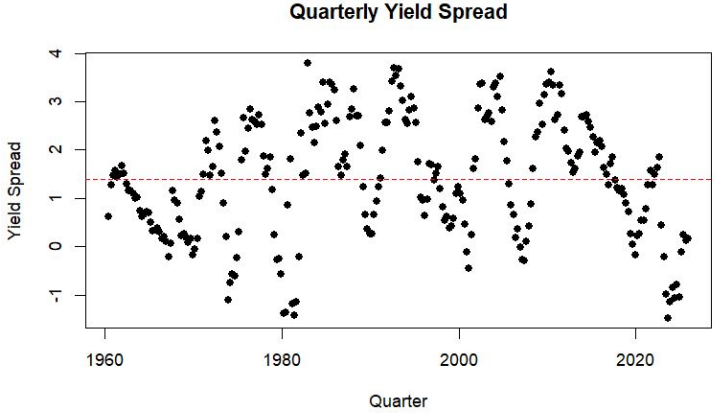
Data contd.


Quarterly GDP Growth
GDP Growth
0.05
0.00
-0.05
1960
1980
2000
2020
Quarter


Quarterly Unemployment Rate
Unemployment Rate
12
10
8
6
4
1960
1980
2000
2020
Quarter


Quarterly Inflation
Inflation
0.04
0.02
0.00
-0.02
1960
1980
2000
2020
Quarter


Quarterly Industrial Production Growth
Industrial Production Growth
0.05
0.00
-0.05
-0.10
1960
1980
2000
2020
Quarter


Quarterly Yield Spread
Yield Spread
4
3
2
1
0
-1
1960
1980
2000
2020
Quarter

Methods

- Frequentist Approach:
  - Multiple Linear Regression

$$\text{gdp_growth}_t = \beta_0 + \beta_{\text{gdp}} \cdot \text{gdp}_{t-1} + \beta_{\text{unrate}} \cdot \text{unrate}_{t-1} + \beta_{\text{inflation}} \cdot \text{inflation}_{t-1} + \beta_{\text{ip}} \cdot \text{ip}_{t-1} + \beta_{\text{spread}} \cdot \text{spread}_{t-1} + \epsilon_t$$

$$\epsilon_t \sim N(0, \sigma^2)$$

- Bayesian Approach
  - Bayesian Linear Regression Standard Model

$$\text{gdp_growth}_t \mid \beta, \sigma \sim N(\mu_t, \sigma)$$

$$\mu_t = \beta_0 + \beta_{\text{gdp}} \cdot \text{gdp}_{t-1} + \beta_{\text{unrate}} \cdot \text{unrate}_{t-1} + \beta_{\text{inflation}} \cdot \text{inflation}_{t-1} + \beta_{\text{ip}} \cdot \text{ip}_{t-1} + \beta_{\text{spread}} \cdot \text{spread}_{t-1}$$

$$\beta_j \sim N(0, 5)$$

$$\sigma \sim \text{Student-t}(\nu = 1, \text{scale} = 5)$$

Methods contd.

- Bayesian Approach contd.
  - Bayesian Linear Regression With t-Distribution

$$\text{gdp_growth}_t \mid \beta, \sigma, \nu \sim \text{Student-t}_\nu(\mu_t, \sigma)$$

$$\mu_t = \beta_0 + \beta_{\text{gdp}} \cdot \text{gdp}_{t-1} + \beta_{\text{unrate}} \cdot \text{unrate}_{t-1} + \beta_{\text{inflation}} \cdot \text{inflation}_{t-1} + \beta_{\text{ip}} \cdot \text{ip}_{t-1} + \beta_{\text{spread}} \cdot \text{spread}_{t-1}$$

$$\beta_j \sim N(0, 5)$$

$$\sigma \sim \text{Half-t}(\nu = 1, \text{scale} = 5)$$

Methods contd.

- Rolling Window Estimation
  - For each window w:

$$\mu_t = \beta_0^{(w)} + \beta_{\text{gdp}}^{(w)} \cdot \text{gdp}_{t-1} + \beta_{\text{unrate}}^{(w)} \cdot \text{unrate}_{t-1} + \beta_{\text{inflation}}^{(w)} \cdot \text{inflation}_{t-1} + \beta_{\text{ip}}^{(w)} \cdot \text{ip}_{t-1} + \beta_{\text{spread}}^{(w)} \cdot \text{spread}_{t-1}$$

$$t \in \{w, \ldots, w + W - 1\}$$

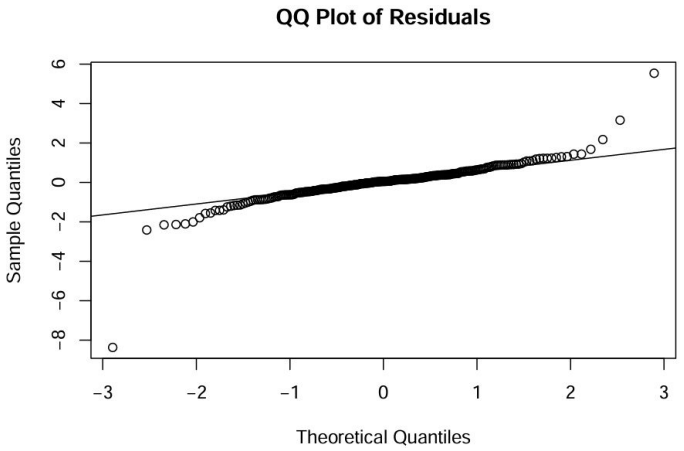
$$\epsilon_t \sim N(0, \sigma_w)$$

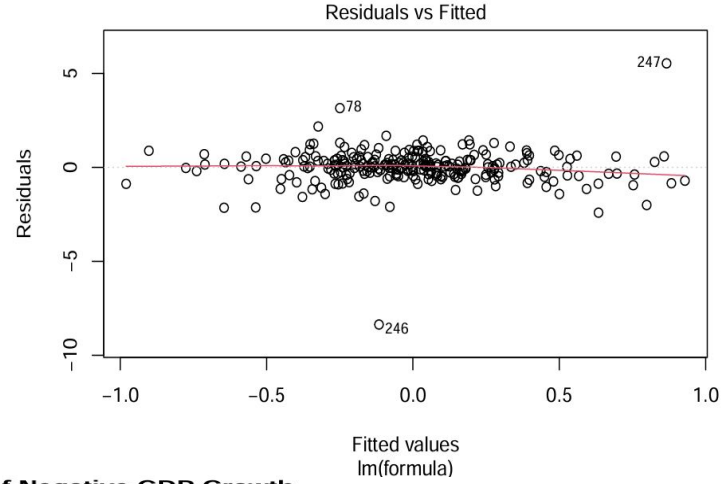
Linear Regression on Interesting Predictors

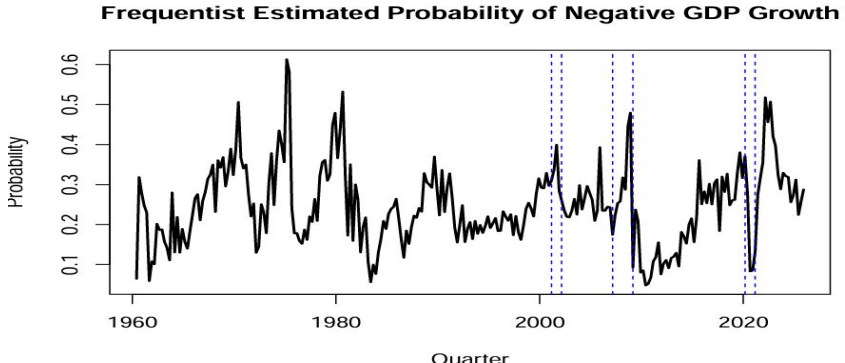
- As we can see by the summary of the frequentist model, most of our predictors are statistically significant since the p-values are below .05 with the expectation of lag_gdp_growth and lag_spread
- In terms of our coefficients, for example, as there is a one standard deviation increase from the previous quarter in inflation, this results in a -.216 increase on that specific quarter's GDP.

```
##
## Call:
## lm(formula = formula, data = df_final_scaled)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -8.3669 -0.3596  0.0522  0.3877  5.5354
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -1.267e-16  5.887e-02   0.000 1.000000
## lag_gdp_growth -1.662e-01  1.052e-01  -1.580 0.115343
## lag_unrate      2.611e-01  7.545e-02   3.461 0.000630 ***
## lag_inflation  -2.160e-01  6.337e-02  -3.408 0.000758 ***
## lag_ip_growth   2.941e-01  1.051e-01   2.800 0.005505 **
## lag_spread     -9.243e-02  7.835e-02  -1.180 0.239173
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9547 on 257 degrees of freedom
## Multiple R-squared:  0.106,  Adjusted R-squared:  0.0886
## F-statistic: 6.094 on 5 and 257 DF,  p-value: 2.359e-05
```

Linear Regression on Interesting Predictors Contd.







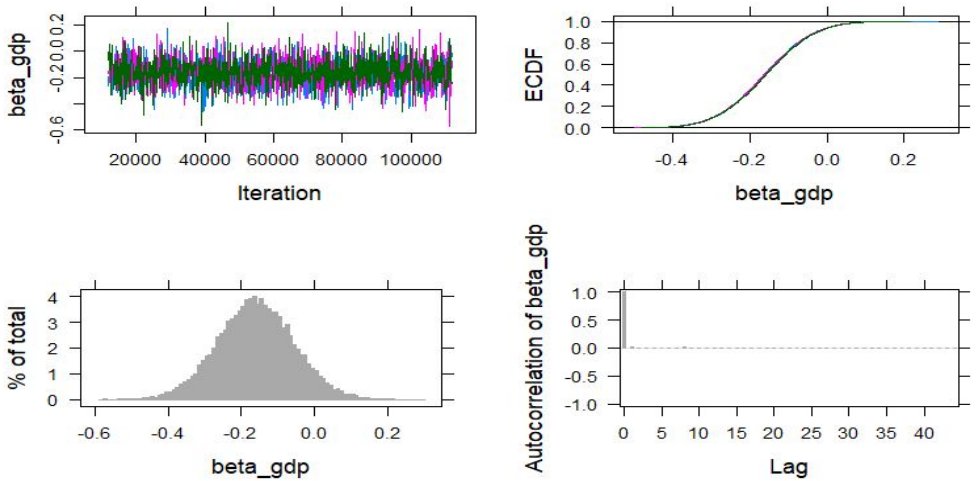
Bayesian Linear Regression

- For both of our Bayesian linear regression portions of the project, our end recession probability plots nearly exactly matched our frequentist results
- We tinkered with the priors to be stronger and the discrepancy remained very small, almost unnoticeable

```
# Model String for Bayesian MLR
modelString <- "
model {
## sampling
for (i in 1:N) {
    mu[i] <- beta_0 +
             beta_gdp * lag_gdp[i] +
             beta_unrate * lag_unrate[i] +
             beta_inflation * lag_inflation[i] +
             beta_ip * lag_ip[i] +
             beta_spread * lag_spread[i]

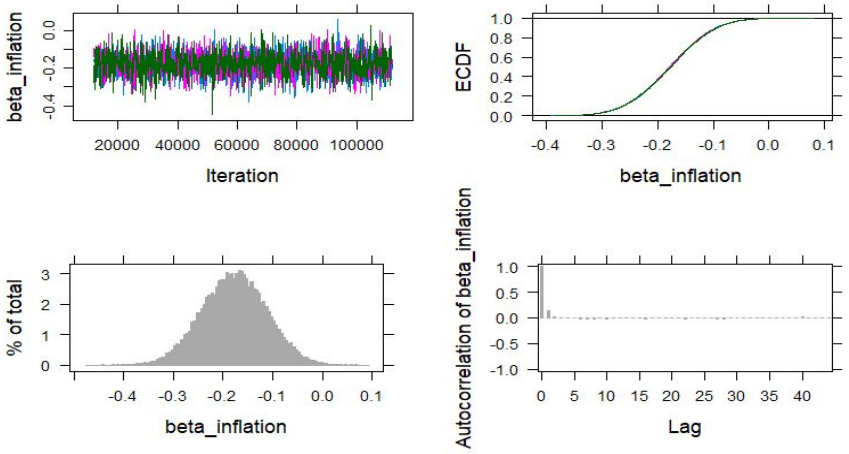
  y[i] ~ dnorm(mu[i], tau)
}

## priors
beta_0           ~ dnorm(mu0, g0)
beta_gdp         ~ dnorm(mu1, g1)
beta_unrate      ~ dnorm(mu2, g2)
beta_inflation ~ dnorm(mu3, g3)
beta_ip          ~ dnorm(mu4, g4)
beta_spread      ~ dnorm(mu5, g5)
sigma ~ dt(mu_sigma, tau_sigma, nu_sigma) T(0, upper_sigma)
tau <- pow(sigma, -2)
}
"
```



Bayesian Linear Regression with Student-t Distribution

- We wanted more interesting results to present, and decided to employ a student-t Bayesian linear regression model to try and remedy the heavy tailed residuals, shown in our regression diagnostic plots
- Our t-model was able to properly capture residual tails but resulted in lower prediction accuracy




```
# Define model string for unscaled t model
modelString_t_unscaled <- "
model {

  for (i in 1:N) {
    mu[i] <- beta_0 +
             beta_gdp * lag_gdp[i] +
             beta_unrate * lag_unrate[i] +
             beta_inflation * lag_inflation[i] +
             beta_ip * lag_ip[i] +
             beta_spread * lag_spread[i]

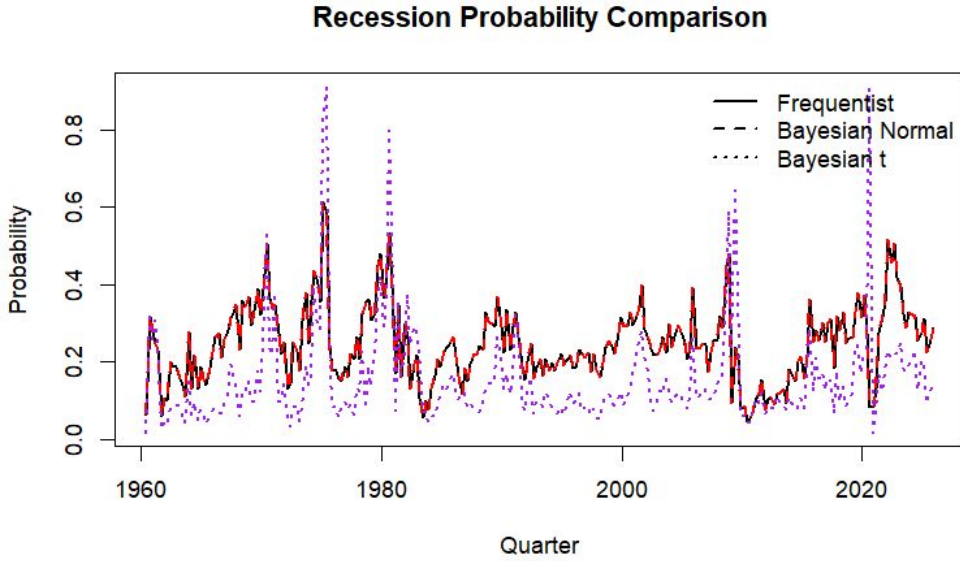
    y[i] ~ dt(mu[i], tau, nu)
  }

  beta_0           ~ dnorm(0, 1/25)
  beta_gdp         ~ dnorm(0, 1/25)
  beta_unrate      ~ dnorm(0, 1/25)
  beta_inflation   ~ dnorm(0, 1/25)
  beta_ip          ~ dnorm(0, 1/25)
  beta_spread      ~ dnorm(0, 1/25)

  sigma ~ dunif(0, 0.1)
  tau <- pow(sigma, -2)

  nu_minus_two ~ dexp(1/30)
  nu <- nu_minus_two + 2
}
"
```

Bayesian Linear Regression with Student-t Distribution contd.



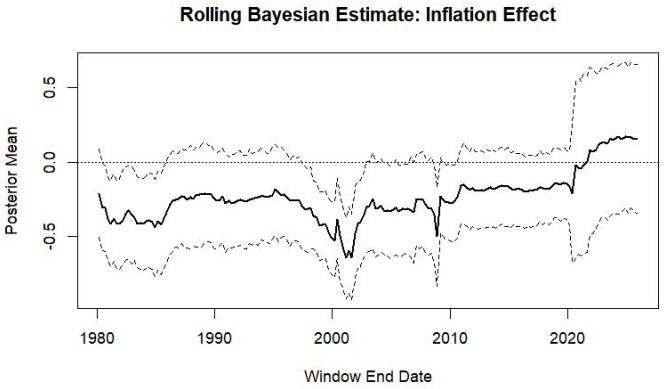
Predictive fit comparison across models

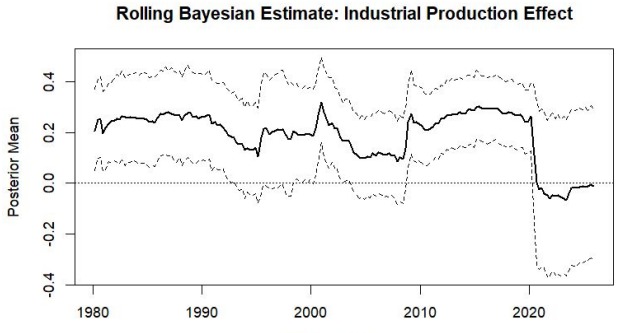
Model	RMSE	MAE
Frequentist	0.00994	0.00604
Bayesian Normal	0.00994	0.00603
Bayesian t	0.34565	0.24248

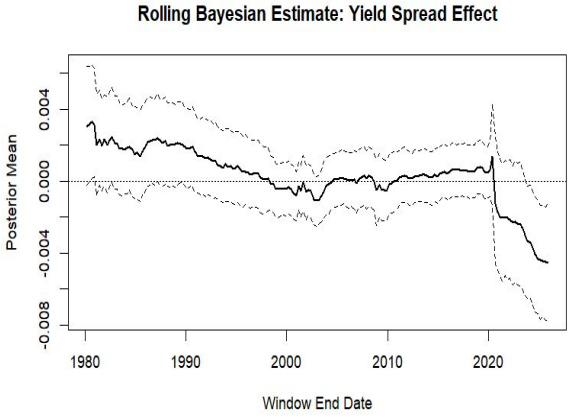
Rolling Window Estimation

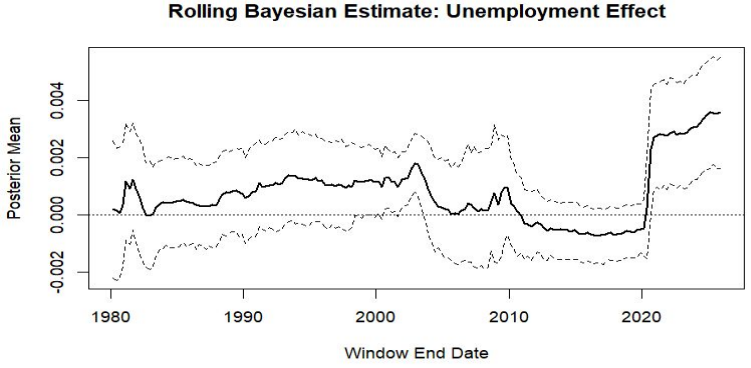
- We performed this analysis to inspect the influence of the lagged predictors over time
- The results are interesting as some began with negative magnitude and ended up positive, the inverse is true
- On the next slide are plots of the upper bound, mean, and lower bound of coefficient values from JAGS simulations with shifted windows in the dataset
- It is interesting to see the balancing that occurs in the coefficient magnitudes over time, despite the regression coefficients not having constraints

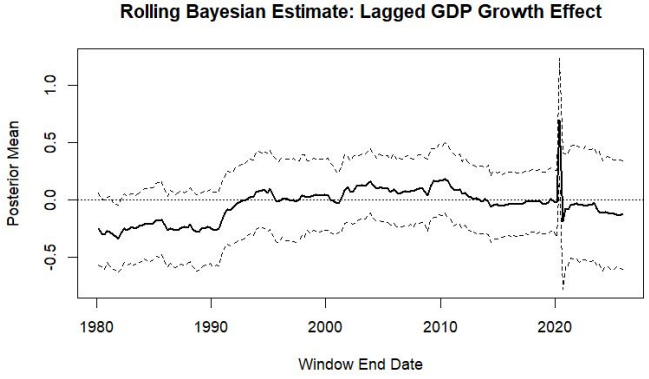
Rolling Window Estimation contd.





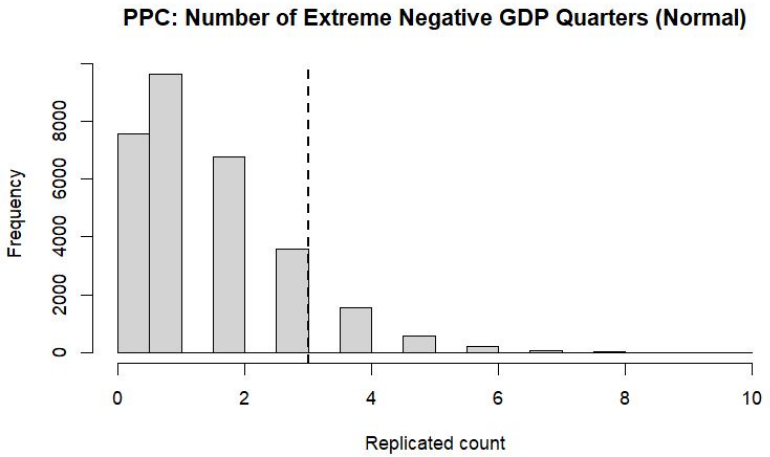


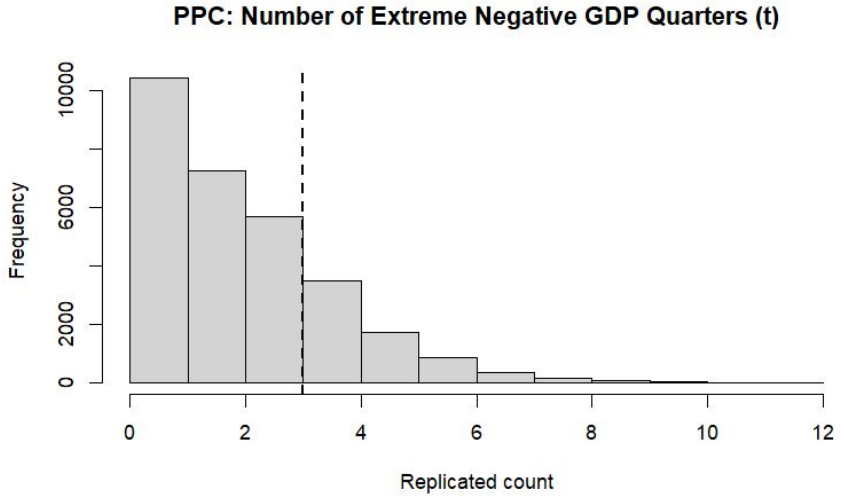




Posterior Predictive Checks

- From our posterior predictive checks, we found both of our Bayesian models had a difficult time predicting extreme negative GPD quarters, which emphasizes the anomalous nature of economic recession



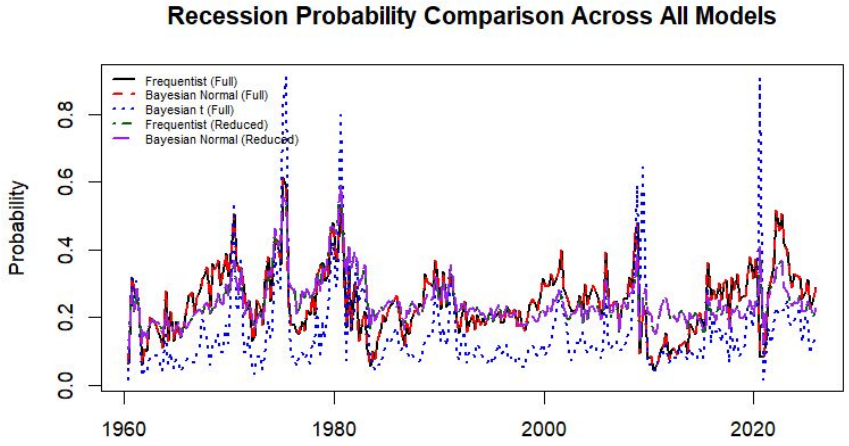


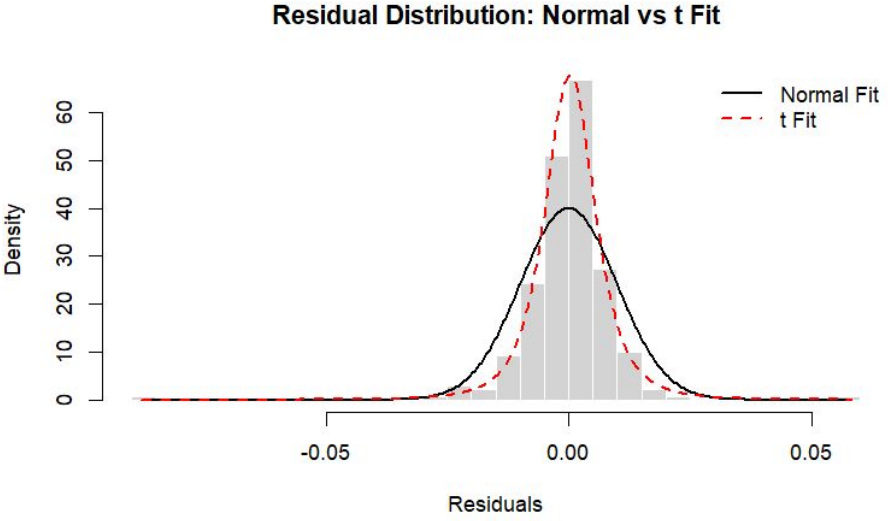
Results and Key Takeaways

- Bayesian Linear Regression is nearly identical to Frequentist Linear Regression if priors are too weak and data is abundant
- T-model captures heavy tails and fits residuals better, hurts prediction
- Influence of predictors on GPD is not consistent through the duration of the dataset
- Extreme cases of recession cause strange behavior in models

Table 12: Posterior predictive checks for Bayesian normal and Bayesian t models

Statistic	Observed	PPP_Normal	PPP_t
mean	0.0073	0.5007	0.5329
sd	0.0105	0.6136	0.2502
min	-0.0820	1.0000	0.9129
prop_negative	0.1217	1.0000	0.9159
n_extreme_neg	3.0000	0.2004	0.4105





Future Considerations

- Different/stronger priors
- Logistic Recession Classification (binary response)
- More/different predictors
- Yearly level data (260+ observations -> 60+ observations)
- Better forecasting approach (ignore future information)

Overall, our project ended up being rough around the edges due to the nuance of time-series/econometric data, but still led to interesting results that agree with Bayesian theory.